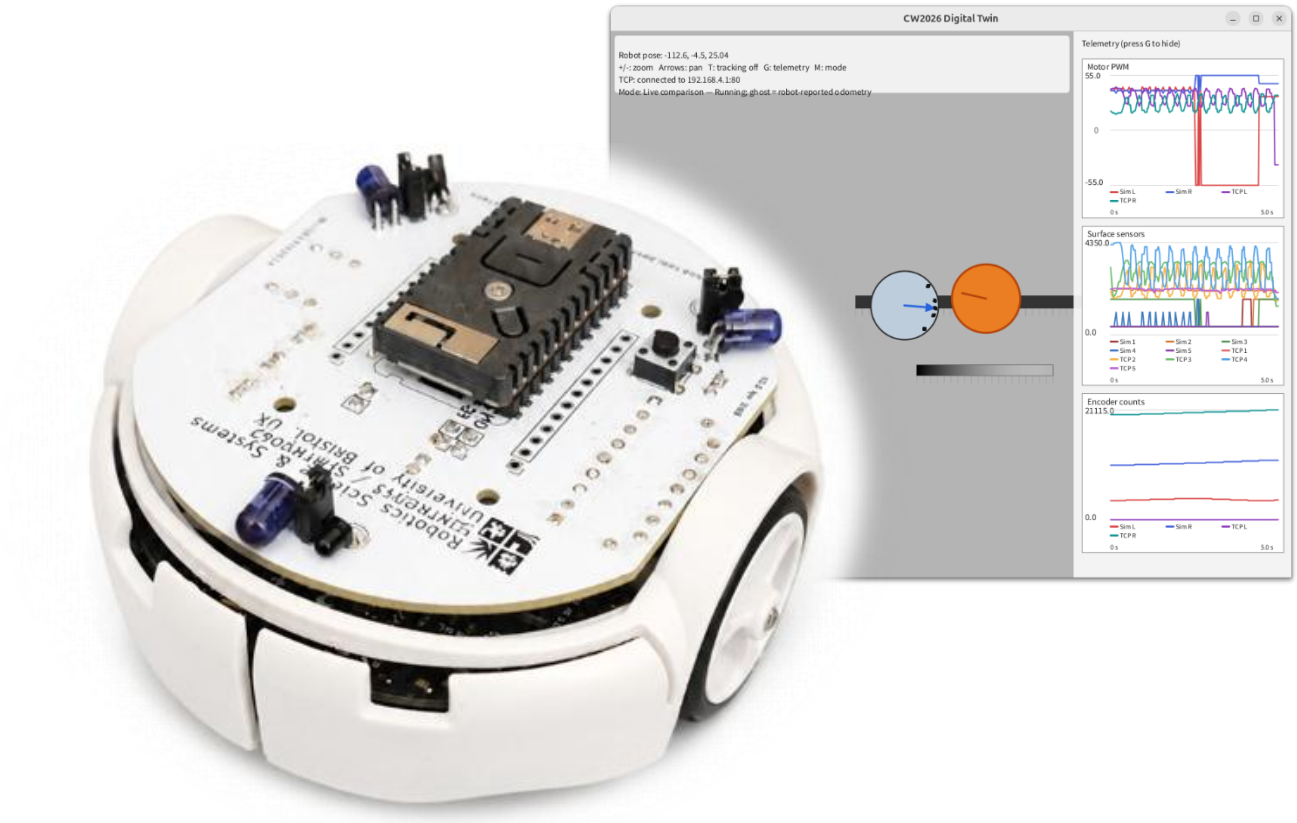


SEMTM0043: Robotics Science & Systems

Academic Year 2026/2027

Unit Director: Dr. Paul O'Dowd paul.odowd@bristol.ac.uk

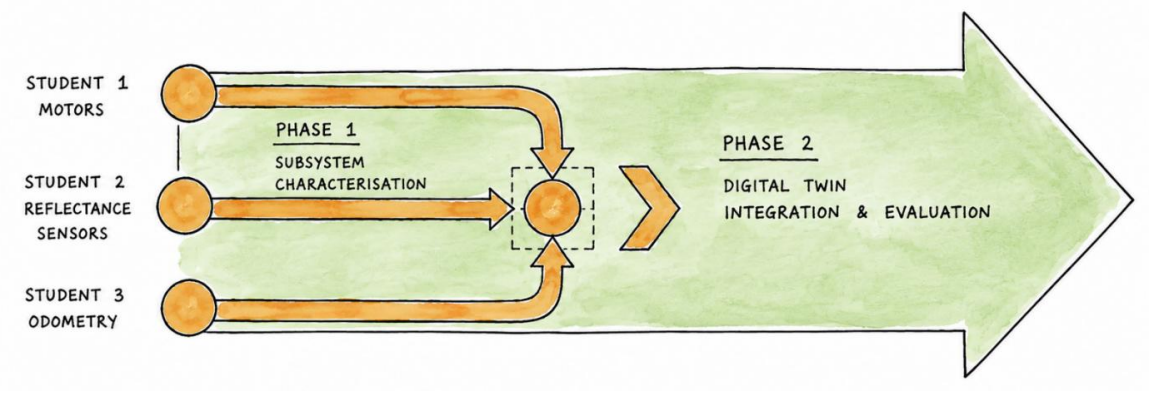
Learning & Assessment in Brief

SEMTM0043 is taught through your practical engagement with a **thematic group project** spanning Teaching Block 1. You will work in groups of 2, 3 or 4 students to investigate and critically discuss a **Digital Twin Robotic System**, with a component of individual work contributing to the wider collaborative group investigation.

A Digital Twin is a simulation counterpart to a real robot. In contemporary robotics, a simulation environment is an essential tool to lower the cost of development in robotics. A simulation can allow for experiments to be conducted in less time than running them with a real robot. A simulation mitigates the costs of physical damages to a robot if something were to go wrong (i.e., a simulation can be quickly reset). However, for any robotics development conducted in simulation to be useful, the simulation must allow for solutions to be transferred and operated in reality – typically referred to as "sim-to-real gap" in state-of-the-art robotics research. Therefore, it is necessary to first investigate the performance characteristics of a real robot and then transfer this understanding into the simulation model. In this way, SEMTM0043 provides two modes of learning and assessment:

- **Phase 1** (Assessment 1 weighted 30%, Individual Basis) focused on studying 1 of 3 subsystems on the real robot. You will need to study and measure the real robot, exercising critical thinking to decide which aspects are most important to understand and model, and how to measure and evaluate them. You are expected to measure and communicate your understanding of a characteristic of either the motors, sensors or odometry subsystem. You will need to consider and report your subsystem in the applied context of your selected project theme. Example project themes are provided in the teaching materials.

- **Phase 2** (Assessment 2 weighted 70%, Group Basis) as a group, to study by comparison the performance of both the real robot and simulated robot for a robotics task of your choosing. You will integrate your measurements and understanding of the real robot into the provided simulation model, so that the operation of the simulation model provides a meaningful and fair comparison to the real robot. For example, you may choose to study a line following behaviour with both the real robot and simulated robot. You will measure both the real robot and simulation, to critically analyse and evaluate how well their performance agrees or disagrees, providing an evidence-supported argument for how and why. It is necessary to work with both parts of this system – it is not sufficient to work only with the real robot, or only with the simulation. The primary question we are interested in is how effectively the real robot and simulation can be used together as a Digital Twin system.



Assessment 1 and 2 both have the final deadline of 1pm Thursday 3rd December 2026. Therefore, you can revise your individual work as you make progress within the group project theme. For example, you may learn something from working as a group on the simulation that improves your understanding of the real robot sub-system. For further details on each assessment specifically, see pages **3** and **6**.

Assessment Specification: Deadline: 1pm Thursday 3rd December 2026

The same deadline applies for both Assessment 1 & 2.

University of Bristol AI-Use**: Category 4 (Integral):** You are permitted, encouraged and supported, to use and to critically reflect on Generative AI tools in your engagement with this unit of study. If you have an ethical objection to Generative AI use, or you would rather study without it, that is also supported throughout the teaching materials provided to you. You must still respect and adhere to any stipulated AI-Use categories for other units of study.

Assessment Submission Requirements:

- Assessment 1, **Individual Basis**, weighted 30% of final unit mark.
  1. You must submit to the SEMTM0043 Blackboard Unit page a technical report (1 A4 page maximum) giving an evidence-based technical characterisation of 1 aspect of a subsystem (motors, sensors, or odometry) of the real robot which you have investigated. Potential aspects of a subsystem to report on are provided as exercises in the teaching materials.
  2. You must provide the code you have independently developed for the StampC3 to measure your subsystem.
  3. You must provide the original data you have independently collected in your investigation, from which your individual report has been produced.
  - The omission of any of (1)(2) or (3) above will result in an awarded mark of 0 (considered *non-submission*).

- Assessment 2, **Groupwork Basis**, weighted 70% of final unit mark.
  1. Your group must submit to the SEMTM0043 Blackboard Unit page a scientific report (3 A4 pages maximum) giving an evidence-based critical evaluation of an experiment conducted with the full Digital Twin Robotic System. This means you must select a task to apply both your real robot and the simulation. From this, you will construct an **evidence-supported argument** for which aspects of the robot's behaviour are most important to model in the digital twin, and to evaluate how and why discrepancies remain between reality and simulation. This will include a critical reflection on the experimental approach undertaken.
  2. You must provide the code you have developed for your robot and digital twin simulation to complete your chosen task (e.g., line following behaviour, conducted both with the real robot and the full simulation).
  3. You must provide the original data you have collected in your investigation, from which your group report has been produced.
  4. The omission of any of (1)(2) or (3) above will result in an awarded mark of 0 (considered *non-submission*).

Marking and Moderation Approach: Both Assessment 1 and Assessment 2 will be marked by the Unit Director following the criteria on Pages 5 and 7. Moderation of marking will be undertaken by a second academic, reviewing a 10% sample of the total submissions for both Assessment 1 & 2 respectively – applying any recommended actions following the University of Bristol Regulations and Code of Practice for Taught Programmes.

Resolving Individual Marks from a Group Mark for Assessment 2: The mark for Assessment 2 reflects the academic quality of the submission produced collectively by the group. To account for individual contributions, the mechanism explained on page **8** is employed to proportionally adjust individual marks from the awarded group mark.

Third Party Libraries: This Unit of study has no restrictions on the use of third party software libraries, either for programming the Pololu 3Pi+ robot or the Digital Twin based in Processing. You must follow the general requirements of Academic Integrity to give attribution and acknowledgement for the use of materials that are not your own. Not giving clear attribution constitutes an attempt to claim credit for someone else's work.

Assessment 1 (Phase 1): Technical Report, 1 A4 Page, Individual Basis (30%)

Working on an individual basis, you should investigate 1 of the 3 identified real robot subsystems: *motors*, *surface reflectance sensors*, or *odometry*. Each subsystem has a set of exercises in the teaching materials to support your progress.

For your selected subsystem, you are required to produce a 1 A4 page technical report that characterises the performance of your subsystem of the real robot in the context of the selected group project and the full Digital Twin system.

Your technical report should identify, evidence and discuss the aspect of your subsystem which is either most challenging to model, or most important to model, within the task your group assigns to the robot for Assessment 2.

For example, a student who has chosen to study the motors subsystem is also working on a group project in which the robot is tasked to scan the surface with its sensors whilst it drives around slowly. In this context, the student decides that the motor characteristic of motor deadband is most important for their project, and to therefore model and evaluate in the simulation & Digital Twin system. Deadband is the focus of their technical report. The technical report will detail a description of deadband, how it was measured, an analysis of what the data can be understood to mean and with what degree of confidence, a reflection on any limitations, and the implications of the acquired understanding for the group project theme, simulation and full Digital Twin system.

A technical report template with recommended sections is provided to you on Blackboard. You are not required to provide a literature review. However, use referencing to any external sources used to support or develop your work (for example, software libraries, datasheets, etc).

Assessment 1 (Phase 1): Technical Report, 1 A4 Page, Individual Basis (30%)

The report should investigate **one subsystem characteristic within the context of the chosen project theme**. Test conditions should reflect its intended use. Where the project involves different operating conditions, the measurements should address the relevant variation.

Grade category	Mark range	What we are looking for
Fail	**0–39%**	Shows little accurate understanding of the chosen characteristic. Measurements are missing or seriously flawed, or their interpretation is substantially incorrect. Testing does not meaningfully address the subsystem context. Conclusions lack support, and the report provides no meaningful connection to Digital Twin modelling.
Fail	**40–49%**	Shows some relevant understanding, but important errors or gaps weaken the explanation. Measurements give insufficient attention to conditions relevant to the subsystem, limiting the conclusions that can be drawn. Possible effects of changing conditions are overlooked or poorly explained, and implications for Digital Twin modelling are unaddressed.
Pass	**50–59%**	Correctly explains the characteristic using measurements under relevant subsystem conditions, including a basic discussion of where operating conditions or outcomes vary. Provides a numerical interpretation and a supported conclusion. Identifies a limitation of the testing and describes a plausible implication for representing the subsystem in a Digital Twin.
Merit	**60–69%**	Selects and justifies relevant test conditions, comparing the measured behaviour through clear, accurate quantitative analysis. Explains how observed variation, or the absence of a clear effect, informs a specific model parameter, feature or assumption. Recognises uncertainty and the limits of the conditions tested.
Distinction	**70–79%**	Uses well-chosen, controlled comparisons to evaluate the characteristic under relevant subsystem conditions. Assesses whether apparent effects are distinguishable from measurement uncertainty and critically evaluates what the evidence supports. Justifies a modelling choice for the intended use and proposes a suitable check of its applicability.
Exceptional Distinction	**80-100%**	Uses particularly informative tests and precise quantitative reasoning to establish how far the findings apply across the subsystems operating conditions. Justifies which contextual effects the Digital Twin should represent or simplify. Explains a consequential limitation or trade-off and identifies the further evidence needed to assess it.

Assessment 2, Phase 2: Scientific Report, 3 A4 Pages, Group Basis (70%)

Assessment 1 focuses on your ability to measure, analyse and communicate a technical understanding of a typical operating characteristic of a robotic subsystem (e.g., motor deadband). In comparison, Assessment 2 asks you to evaluate the Digital Twin system as a whole, within a project theme of your choosing.

Assessment 2 therefore shifts the focus of investigation from a subsystem studied under controlled conditions to a complete robotic system operating within a more complex task. This requires you to develop your critical thinking and scientific investigation skills further. Given your chosen project theme, you will need to conceive, design and conduct an experiment that identifies and evaluates a meaningful aspect of the relationship between the behaviour of the real robot and its simulation.

Working in a group of 2, 3 or 4 students, you should design and conduct an experiment that allows you to analyse and evaluate the capabilities and limitations of the Digital Twin system. For example, you might investigate:

- how accurately and precisely the simulation represents the behaviour of the real robot, and what observed discrepancies reveal about the simulation models employed; or
- the extent to which the simulation can predict the behaviour of the physical robot, and what comparison between the two can reveal about the robot itself.

Your aim is to critically examine the relationship between the digital twin simulation and the physical robot. Differences between the two should not simply be regarded as errors to eliminate: they provide evidence that can be used to investigate modelling assumptions, limitations, unmodelled effects and differences in behaviour.

We are not looking for a large volume of code or a polished solution to a particular robotics problem. Instead, approach the project as a scientist: ask a meaningful question, design an appropriate experiment, take robust measurements, analyse the resulting evidence, and use that evidence to develop and defend your conclusions.

A strong piece of work will draw evidence from multiple measurements or perspectives; collect data in a robust and scientifically fair manner; gather sufficient evidence to meaningfully investigate uncertainty and ambiguity; and use appropriate comparisons to establish what can reasonably be concluded, and with what confidence. Your report should also provide sufficient technical and methodological detail for the experiment to be reproduced in principle.

A scientific report template is provided on Blackboard. Your report should communicate and discuss:

- the task selected for your robotic system, and why it provides a useful or interesting context for investigation;
- the experimental methodology you designed and used;
- the results obtained, and what those results provide evidence for; and
- a critical discussion of the experiment as a whole, including relevant technical limitations, uncertainties, assumptions and aspects of the development process.

Importantly, the intended outcome of this assessment is not a perfect solution to a robotics problem, nor necessarily a digital twin that reproduces the physical robot perfectly. A close correspondence between simulation and reality can be a meaningful result, but only where your experiment provides sufficient evidence to demonstrate it. Equally, discrepancies between the two are not necessarily failures: when investigated carefully, they can provide valuable evidence about the assumptions and limitations of the simulation, the behaviour of the physical robot, or both.

You should therefore choose a task and design an experiment that provides sufficient opportunity to investigate the complexities and challenges of a digital twin robotic system. In doing so, you will demonstrate your ability to design an investigation that is both scientifically rigorous and sufficiently representative of the behaviour you wish to understand.

Assessment 2, Phase 2: Scientific Report, 3 A4 Pages, Group Basis (70%)				
Grade category	**Mark band**	**Technical Approach & Experimental Method (40% weight)** Appropriateness of implementation, measurements, controls and comparison	**Analysis & Critical Evaluation (40% weight)** Correct analysis, interpretation, strength of evidence and implications	**Scientific Communication (20% weight)** Clear, concise and traceable presentation of the investigation
Fail	**0–39%**	The investigation lacks a coherent question or a suitable technical approach. Evidence from the real robot or simulation is absent, insufficient or obtained through a fundamentally unsuitable method. Serious problems with measurements, controls or comparison prevent a defensible evaluation. Essential implementation or experimental details are missing or incorrect.	Provides little valid analysis or supported understanding. Results are absent, substantially incorrect or presented without a meaningful robot–simulation comparison. Conclusions are missing, contradict the evidence or make claims the experiment cannot support. Shows little understanding of relevant model assumptions, measurement limitations or the significance of the findings.	The report is very difficult to follow. Its question, method, evidence or conclusions cannot be identified reliably. Missing or unreadable figures, incorrect labels or units, or unclear distinctions between measurements and simulation prevent assessment of the work. Essential context or attribution is absent.
Fail	**40–49%**	Identifies a relevant task and shows some appropriate technical work, but the question or method is insufficiently defined. Evidence from the robot and simulation supports only an incomplete or unreliable comparison. Important weaknesses in conditions, measurements, controls or implementation undermine the evaluation. Missing technical details prevent a reliable judgement of how the investigation was conducted.	Identifies some relevant observations, but analysis contains important errors or is insufficient to support the conclusions. Comparisons are weak, and explanations are largely asserted or descriptive. Important uncertainty, limitations or alternative explanations are overlooked. The report does not establish clearly what the evidence shows about the relationship between the robot and simulation.	Contains relevant material, but weak organisation or unclear writing obscures the investigation. Figures, captions or technical descriptions omit information needed to interpret the results. Links between claims and evidence are difficult to follow. Space is poorly allocated, or sources and reused material are not identified consistently.
Pass	**50–59%**	Defines a focused, answerable question about the robot–simulation relationship and uses a basically sound technical approach. Collects relevant evidence from both systems under sufficiently comparable conditions. Identifies the main measurements, controls and repetitions, with enough detail to understand the essential evaluation. Describes any model changes or calibration and the data used. Some choices are weakly justified, or methodological limitations reduce confidence.	Processes and compares results using basically correct, appropriate quantitative measures. Draws a supported conclusion that addresses the question and relates the findings to the twin's intended use. Identifies a relevant source of uncertainty or limitation and explains its effect on the conclusion. Shows basic understanding of relevant behaviour or model assumptions, although explanations and critical evaluation remain limited.	Communicates the question, method, main results and conclusion in an understandable three-page report. Figures are readable, with appropriate labels, units and captions. Measurements and simulated results are distinguishable. Provides sufficient technical context and identifies sources and reused material. Some repetition, uneven detail or weak links between text and figures reduce clarity.
Merit	**60–69%**	Uses an appropriate implementation and a well-defined experiment, with clear reasons for the main choices. Selects relevant conditions and measures, controls important influences and uses appropriate repetition or sampling. Addresses principal measurement limitations and clearly separates calibration data from evaluation data. Reports settings and procedures clearly enough to repeat the essential comparison. Recognises remaining design limitations.	Applies appropriate analysis correctly and draws clear, supported conclusions from quantitative comparisons. Examines variability or uncertainty and explains its significance for the findings. Relates observed agreement or differences to relevant model assumptions or subsystem effects. Considers plausible alternative explanations, distinguishes supported inferences from untested explanations, and evaluates the twin's usefulness for the stated task and conditions.	Presents a clear, coherent scientific account with a logical progression from question and method to evidence and conclusions. Well-chosen figures, accurate units and informative captions support the argument. Technical detail is concise and relevant. Sources are identifiable, and the reader can distinguish observations, predictions and interpretation. Uses the available space effectively.
Distinction	**70–79%**	Shows strong technical and experimental judgement through well-justified choices. The method provides a fair, informative test under conditions relevant to the question. Controls, repetition and reference measurements address important risks of misleading comparison. Where fitting occurs, clearly separates calibration from independent evaluation. Explains consequential design trade-offs and provides a credible basis for reproducing the essential evaluation, including relevant model settings.	Builds a convincing, evidence-based account of the robot–simulation relationship. Uses suitable analysis to evaluate plausible explanations and assess whether discrepancies matter for the intended use. Critically examines the method's informativeness, uncertainty and representativeness. Clearly distinguishes what is established from what remains unresolved, and justifies the conditions within which conclusions about prediction, modelling or control can be applied.	Develops a precise, economical technical argument within three pages. Purposeful organisation and carefully selected evidence make the reasoning easy to assess. Figures and text work together to communicate comparisons, uncertainty and limitations. Technical details and attribution allow the reader to trace the key claims without unnecessary repetition or reliance on unstated assumptions.
Exceptional distinction	**80–100%**	Shows exceptional engineering judgement in a focused, effective investigation. Selects particularly informative conditions or measurements to distinguish influential factors, addresses consequential measurement or comparison errors, and tests robustness under relevant variations. Technical choices balance accuracy, complexity and practical constraints with precise justification. The essential evaluation is reproducible, and any model fitting is clearly separated from the evidence used to assess predictive performance.	Synthesises results into insightful, well-supported understanding that provides a clear basis for decisions about modelling, evaluation or use of the system. Resolves a meaningful uncertainty, establishes why it remains unresolved, or exposes a consequential limit or trade-off. Weighs the strength of competing explanations with exceptional judgement and justifies both the scope of the findings and the further evidence needed.	Communicates the investigation with exceptional clarity and economy. The question, decisive evidence, reasoning and limits are immediately apparent and remain easy to trace. Figures, captions and concise technical explanations make the important comparisons readily assessable. Precise terminology, transparent attribution and an integrated presentation support critical reading throughout.

Resolving Individual Marks from an awarded Group Mark: Equity Share

For Assessment 2 only, your team will consists of 2, 3 or 4 students. If everyone contributes equally, each team member has an equal equity share (e.g., [50%, 50%], [33.33%, 33.33%, 33.33%], [25%, 25%,25%, 25%]. During the project, equity shares will be considered across 4 activities:

1. **Two Supervision meetings**: You will have 2 supervision meetings, scheduled to occur across weeks 7,8 (meeting 1) and 9,10 (meeting 2). If you do not attend a meeting, and your team members cannot agree your contribution on your behalf, it will be recorded as 0%.
2. **Two Peer-Assessments**: You will provide a confidential assessment of yourself and your team members in week 6 and week 11. If you do not submit a peer-assessment, your peer-assessment will be assumed to be equal contribution between all team members.

An equity share represents your share of responsibility for the team's achievement, not simply the amount of work completed. You should consider contributions to include technical work, ideas and decision-making, experimentation and analysis, problem solving, integration, and reliable participation in the team's activities.

When assessing equity, the agreed equity shares must always total 100%. For example, in a team of 3:

Student A = 35%, Student B = 35%, Student C = 30%: Total = 100%

You should be prepared to discuss an equity share with evidence of contribution. The first peer-assessment and subsequent supervision meetings are intended to help identify imbalances while there is still time to address them. The complete pattern of equity shares across the project will provide evidence when determining individual contribution to the final mark awarded to the work produced as a group submitted for Assessment 2.

The four records of equity share will be used together to determine how significantly each individual's contributions differed from an ideal equal contribution. This difference is used to adjust the awarded group mark to an individual mark. The below table illustrates an example outcome against a group mark of 66%.

(%)	Peer-Assessment 1 Averages	Meeting 1 Agreed Contributions	Meeting 2 Agreed Contributions	Peer-Assessment 2 Averages	4 Records Averaged	Difference from Ideal of 33.33%	**Individual Mark from Group mark 66%**
Student A	33.3	20%	30%	30%	28.325	-5.0	61%
Student B	33.3	40%	30%	33%	34.075	+0.7	67%
Student C	33.3	40%	40%	37%	37.575	+4.2	70%

If the span of individual marks differ by more than 10%, if there is no agreement on marks, or if a student should fail as a result, a viva process will be held to determine the final outcome.